I/O Devices and Device Controllers

1 I/O Devices

Basic Idea

I/O devices connect a computer to input, output, storage, communication, or user interaction.

- Examples: disks, keyboards, mice, displays, network cards, printers, scanners, cameras, and buses.
- Devices differ by speed, data format, access method, and control mechanism.
- The OS hides many hardware details behind general device models.

2 Block Devices

Definition

A **block device** stores data in fixed-size addressable blocks. Transfers happen one or more complete blocks at a time.

Property	Meaning
Block size	Usually 512 bytes to 65,536 bytes.
Block address	Each block has a number or address.
Independent access	Any block can be read or written separately.
Examples	Hard disks, SSDs, USB sticks, and Blu-ray discs.

Key Point

Block devices support direct access to a named block.

3 Character Devices

Definition

A **character device** sends or receives a byte stream without addressable blocks.

Property	Meaning
Stream-oriented	Data is processed in sequence.
No block address	The OS cannot choose block N.
No seek operation	The OS cannot jump to an arbitrary position.
Examples	Keyboards, mice, printers, network interfaces, and scanners.

Key Point

Character devices are used as streams, not random-access storage.

4 Limits of the Classification

Imperfect Model

The block-device and character-device model is useful, but some devices do not fit cleanly.

Device	Why it is special
Clock	Produces timer interrupts, not blocks or byte streams.
Memory-mapped screen	Acts like memory plus display hardware.
Touch screen	Combines display and input events.
Tape drive	Has blocks, but reaching block N is sequential and slow.

Practical Use

File systems can target abstract block devices while drivers handle hardware-specific details.

5 Device Data Rates

The I/O subsystem must handle both very slow human input and very fast buses or networks.

Device, network, or bus	Typical data rate	Category
Keyboard	10 bytes/sec	Human input
Mouse	100 bytes/sec	Human input
56K modem	7 KB/sec	Communication
Scanner at 300 dpi	1 MB/sec	Input
Digital camcorder	3.5 MB/sec	Media input
4x Blu-ray disc	18 MB/sec	Block storage
802.11n wireless	37.5 MB/sec	Network
USB 2.0	60 MB/sec	Bus
FireWire 800	100 MB/sec	Bus
Gigabit Ethernet	125 MB/sec	Network
SATA 3 disk drive	600 MB/sec	Storage interface
USB 3.0	625 MB/sec	Bus
SCSI Ultra 5 bus	640 MB/sec	Bus
Single-lane PCIe 3.0 bus	985 MB/sec	Bus
Thunderbolt 2 bus	2.5 GB/sec	Bus
SONET OC-768 network	5 GB/sec	Network

Speed Range

I/O rates range from bytes/sec to GB/sec.

6 Device Controllers

Definition

A **device controller**, or **adapter**, is the electronic component that controls an I/O device.

Part	Role
Mechanical component	The physical device.
Electronic component	The controller or adapter.
Location	Motherboard chip or expansion card, such as PCIe.
Scope	One controller may manage multiple similar devices.

Modularity

Standard interfaces let devices and controllers work together without exposing every hardware detail to the OS.

Common interfaces: SATA, SCSI, USB, Thunderbolt, and FireWire.

7 Controller Responsibilities

Main Job

The controller converts raw device signals into blocks, bytes, or events the OS can use.

For a disk, a sector usually has this structure:

Preamble → **Sector data** → **Checksum/ECC**

Step	Controller action
Receive	Accept the raw bit stream.
Buffer	Assemble bits into bytes or blocks.
Validate	Check checksum or ECC.
Correct	Fix recoverable errors.
Transfer	Copy verified data to main memory.

Key Point

The OS talks to the controller; the controller handles electrical and timing details.

8 Display Controller Example

Without a controller	With a controller
Software controls pixels manually.	The OS sets display parameters and writes display data.
Code depends on device details.	The controller converts memory data into display signals.
Portability is poor.	One software model can support many displays.

Historical Note

LCD screens replaced CRT monitors because they are thinner, lighter, less fragile, and more power efficient.

9 Worked Example

Problem

A disk sector has 512 bytes of payload. How many payload bits must the controller assemble?

Solution

$$512 \text{ bytes} \times 8 \frac{\text{bits}}{\text{byte}} = 4096 \text{ bits}.$$

Answer

The controller assembles 4096 payload bits, excluding preamble, checksum, and ECC bits.